Round 3: Deep Third-Cumulant Compression in a Random ReLU MLP

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21 September 2026

Bottom line. I do not see a deterministic exact representation that beats the two-leg/two-slice-product structure down to ≤ 1.2 u per source-layer. The shared-propagator and Hermite directions reduce to the same symmetric-square coefficient space that your measurements already show is large. However, there is one genuinely different route worth testing: **random coding over the birth/source index**, not over hubs or tensor coordinates. It replaces m old sources by $q = 2$–3 pooled matrix pairs and is exactly unbiased for the desired sum conditional on the fixed network. Three pooled channels cost 6.12 u per layer, independent of m. Its only unknown is coding variance. A second, biased route—fitting each source to an effective diagonal third leg—costs 1.02 u per source-layer and directly exploits the repeated T_b legs, but I expect it to fail unless a surprisingly large part of the dense birth leg is query-null.

1. New proposal: cubic source-index coding

Define the cubic source functional

$$\Phi(T, A) = (T \circ T)A^\top + 2(T \circ A)T^\top, \qquad D_{21} = \frac{1}{3}\sum_b \Phi(T_b, A_b).$$

Here A_b denotes the transported third leg $\widetilde{A}_b$.

[Derived] Φ is homogeneous cubic. Let ζ_b be independent real codes satisfying

$$\mathbb{E}\zeta = 0, \qquad \mathbb{E}\zeta^3 = 1.$$

For one coding channel define

$$T^{(q)} = \sum_{b\in\text{old}} \zeta_{qb} T_b, \qquad A^{(q)} = \sum_{b\in\text{old}} \zeta_{qb} A_b.$$

Expanding $\Phi(T^{(q)}, A^{(q)})$ gives terms with code factor $\zeta_b\zeta_c\zeta_d$. Independence and zero mean kill every index pattern except $b = c = d$. Therefore, conditional on the realized network,

$$\boxed{\mathbb{E}_\zeta\, \Phi(T^{(q)}, A^{(q)}) = \sum_b \Phi(T_b, A_b).}$$

This is not an annealed/even closure: every pooled matrix is a linear combination of the actual quenched propagators, and the output remains odd under the relevant weight sign flip.

A convenient two-point real code is

$$\zeta = \varphi \quad \text{with probability } 0.276393, \qquad \zeta = -\varphi^{-1} \quad \text{otherwise,}$$

where $\varphi = (1 + \sqrt{5})/2$. It has $E\zeta = 0$, $E\zeta^2 = E\zeta^3 = 1$, $E\zeta^6 = 5$; among two-point real codes with the first and third moments fixed this minimizes the sixth-moment variance proxy.

State and update. Per coding channel keep only two dense matrices $(T^{(q)}, A^{(q)})$, i.e. $2n^2$ floats. All old sources see the same next map M, so

$$T^{(q)} \leftarrow MT^{(q)}, \qquad A^{(q)} \leftarrow MA^{(q)}.$$

When a source crosses the age gate, add $\zeta_{qb}T_b$ and $\zeta_{qb}A_b$ to that channel. No source identity is needed afterward.

Cost. One channel uses two level-5 Strassen products for transport and two for the slice:

$$2(0.51) + 2(0.51) = \boxed{2.04\,\text{u/layer}}.$$

Thus $q = 2$ costs 4.08 u/layer and $q = 3$ costs 6.12 u/layer, below the requested 8 u old-tier ceiling. State size is $4n^2$ or $6n^2$ floats (16 or 24 MB in fp32). Applied to the whole K3 history rather than only the old tier, three channels

cost about $16 \times 6.12 = 97.9$ u for all source transport/slicing, which is in the right ballpark for a 133–165 u total method after the covariance/closure bill.

Variance control. Do not code the cheap partial-trace piece. For arbitrary X, Y, write

$$X \circ Y = \mathbf{1} m_{XY}^\top + F_{XY}, \qquad (m_{XY})_j = n^{-1} \sum_i X_{ij} Y_{ij}.$$

The rank-1 part of each coded Φ is cubic and can be subtracted exactly in $O(n^2)$, while the exact sum of the per-source thin parts is added back. The random estimator then acts only on the expensive centered fluctuation F, avoiding variance from a much larger component you already know how to compute.

A useful hybrid under the same budget is to keep p troublesome sources exactly and use q coded pools for the rest. Each exact source or coded pool costs about 2.04 u/layer, so $p + q = 3$ costs 6.12 u/layer. Examples: $(p, q) = (0, 3), (1, 2), (2, 1)$.

[**Measure-this**] **First falsifier (the experiment I would run first).** At layers 10 and 14, use the stored dense old legs and compute the exact centered old-tier sum once. Generate 16–32 independent skew-code channels; each sample requires only two dense slice products because no transport is needed for this test. From those samples estimate the relative RMS error of averages of $q = 1, 2, 3$, and also test $(p, q) = (2, 1)$ after peeling the two sources with largest centered contribution norm. Kill the idea if the best configuration with $p + q \le 3$ is above 2% at either layer. This is roughly 32–64 dense products and should fit comfortably within an hour in NumPy.

Important limitation. Exact compression with $q < m$ coded channels is impossible for arbitrary sources. Moment matching would require a rank-q representation of the order-3 superdiagonal tensor $\sum_{b=1}^m e_b^{\otimes 3}$, whose CP rank is m. Therefore this proposal lives or dies entirely on the *actual contracted cross-source interference* being small after the thin control variate. It is an approximation, but a new one: it sketches the source index ($m \lesssim 10$), not the hub index ($n = 1024$), so item 6 does not test it.

2. Second proposal: fit an effective diagonal third leg

[**Derived**] The two slice products are the Fréchet derivative of one cubic matrix map. Define

$$F(T) = (T \circ T) T^\top.$$

Then

$$DF_T[A] = (T \circ T) A^\top + 2(T \circ A) T^\top = \Phi(T, A).$$

Because $A = T A_b$, the expensive second leg is a tangent generated by the birth matrix. If the birth matrix were diagonal, $A = T \operatorname{diag}(d)$ and the two slice terms collapse exactly:

$$\frac{1}{3} \Phi(T, T \operatorname{diag} d) = (T \circ T) \operatorname{diag}(d) T^\top.$$

So a diagonal birth surrogate needs only one transported matrix and one slice product.

State/update/cost. Keep (T_b, d_b): $n^2 + n$ numbers. Update $T_b \leftarrow M T_b$; d_b is fixed. Cost is

$$0.51 \text{ u (transport)} + 0.51 \text{ u (slice)} = \boxed{1.02 \text{ u/source-layer}}.$$

It preserves the odd full-rank fluctuation because $T \circ T$ is still formed from the realized propagator; it does not replace it by an expectation.

The nontrivial version is not $d = \operatorname{diag} A_b$, but a *query-optimal* d. At the age gate let $X = T \circ T$ and let $S = \Phi(T, A)/3$ be the exact source query already available from the young calculation. Fit

$$\min_d \| S - X \operatorname{diag}(d) T^\top \|_F^2.$$

Its normal matrix is

$$G = (X^\top X) \circ (T^\top T),$$

and the right-hand side is $h_r = x_r^\top S t_r$. A full fit costs about 1.70 u once per source (two Gram products 1.02, one ST product 0.51, Cholesky about 0.17). A diagonal-Gram approximation reduces the one-time fit to about 0.51 u.

[**Conjecture**] I give this a lower prior than source coding. Dense off-diagonal entries of A_b are individually $O(n^{-1/2})$ but there are $O(n^2)$ of them, so their Frobenius energy is not asymptotically negligible. The only hope is that most of that energy is null in the specific future query family and can be absorbed into n effective hub weights.

[**Measure-this**] Fit d for each source at age 4 using the exact current query, then evaluate the surrogate at the next 1, 2 and 4 stored layers without refitting. Kill it if the median error exceeds 2% after one step, or if any source needed by item 5 drifts above about 3% within two steps. This directly tests the off-slice failure mode that killed slice recursion.

3. Why the requested algebraic directions do not otherwise open a door

(i) Sharing the squared propagator. **[Derived]** For $T_b = T_c X_b$, each row obeys

$$(T_c X_b)_{i:}^{\odot 2} = \mathrm{svec}(t_{c,i} t_{c,i}^\top)^\top \, Q(X_b),$$

where $Q(X_b)$ contains the symmetric Kronecker squares of the columns of X_b. Thus the exact shared object is the symmetric-square lift of T_c, with width

$$N_2 = n(n+1)/2 \simeq 5.25 \times 10^5.$$

The lifted matrix has size $n \times N_2 = \Theta(n^3)$ and contracting it with an $N_2 \times n$ object costs $\Theta(n^4)$ (about $N_2/n \approx 512$ dense-matmul units). Compressing this lift is precisely a Tucker/separation-rank approximation; your measured $r \approx 250$ requirement is the obstruction. So the identity exists, but it makes the wall explicit rather than removing it.

(ii) Hermite/Wick in the current Gaussian weight row. **[Derived]** This expansion is already finite and exact. For a fixed incoming tensor, with g_i a Gaussian row and K_c the symmetric matrix slice,

$$q_{ic} = g_i^\top K_c g_i = \mathrm{tr}(\Sigma K_c) + \langle K_c,\, g_i g_i^\top - \Sigma \rangle.$$

The first term is your cheap thin/trace piece; the entire hard part is exactly second Wiener chaos. There are no higher Hermite orders to truncate. The coefficient operator maps $\mathrm{Sym}^2(\mathbb{R}^n)$ (dimension N_2) to $\mathbb{R}^n$. A cheap chaos basis therefore requires low separation rank of that operator, which is the same rank phenomenon already measured by the pooled Tucker/vector-metric experiments. Low chaos *order* is not low coefficient *rank*.

(iii) Backward linear response. **[Derived]** Pulling a quadratic query backward through a linear map preserves its rank-one quadratic form, but the target is a full $n \times n$ D21 family at every insertion layer (or, after composing with the final mean, n output coordinates with a dense quenched sensitivity). Without a low-rank final sensitivity, backward propagation merely exchanges the forward source triangle for a backward query triangle. The Fréchet identity above is the local manifestation: a generic tangent A still requires the second propagated matrix. A decisive check would be a randomized SVD of the Gaussian-chain Jacobian $\mathrm{vec}(D21_l) \mapsto \delta\mu_{16}$; unless $> 99.96\%$ of its energy lies in roughly $r \le 8$ modes, it cannot beat the coded-pool cost. I would not prioritize this over the coding test.

(iv) Cavity/TAP/turbulence/tensor-network closures. **[Conjecture]** Standard moment closures are useful for ensemble statistics but discard precisely the quenched odd phase that parity says is essential. A temporal tensor-network reorder also has a three-index separator; compressing that separator returns to the measured pooled rank ~ 250. I see no distinct $O(n^2)$ state here after items 7–8.

4. What I think the leaders could be doing

[Conjecture] If the source-index coding experiment succeeds with two or three channels, it is the first mechanism in these three rounds that naturally explains both signatures: (1) cost nearly independent of the number of old births, and (2) preservation of the actual odd, full-rank fluctuation rather than its zero conditional mean. A practical leader implementation could keep the newest one or two sources exact and pool the rest into one or two skew-coded channels; four total channels cost about 8.16 u/layer for all K3 work and land naturally in the observed 133–165 u range once the rest of the chain is added.

The remaining 1.3x accuracy advantage over V29 does not require a full K4 tensor. After halving K3 cost, even a local/one-step augmented correction costing one extra Strassen product per layer is only $0.51 \times 16 \approx 8.2$ u total. That is the scale I would allocate to the extra rung before considering any transported n^4 state.

If coding fails: then I cannot beat the practical two-product/tangent floor with the allowed operations. The strongest argument is that the hard query is the action of a high-rank symmetric-square coefficient operator on a realized second-chaos feature, while the repeated-leg structure only removes the second slice product when the third leg is effectively diagonal. Your measurements rule out every known low-rank replacement of that coefficient operator. In that case the only credible bill-halving approximation is exactly the query-optimal diagonal-third-leg test above; if that also drifts above 2%, I would regard the leader mechanism as requiring a qualitatively different approximation not represented in the current K3 source algebra.